Trainings

General Information

⚠ CAUTION ⚠

The 15L engine presently has five different cylinder liner designs available. It is important to identify the liner being removed to properly select the replacement liner.

Current 15L replacement cylinder liner designs for service:

	Antipolishing Ring Liner	Cylinder Liner Outside Diameter	Liner Midstop Shim	Flat Top or Groove Top	Identification Band Machined Around the Bottom Outside Diameter of Liner
1	Yes	150 mm [5.906 inch]	Captive Shim	Flat Top	2
2	Yes	152 mm [5.984 inch]	Service Shim	Flat Top	2
3	No	150 mm [5.906 inch]	Captive Shim	Flat Top	1
4	No	152 mm [5.984 inch]	Service Shim	Flat Top	1
5	No	150 mm [5.906 inch]	No Shim	Groove Top	0

The design used in a particular engine is dependent on the engine's production vintage.

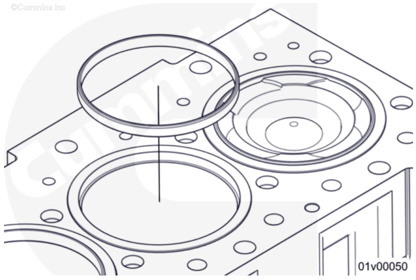
The cylinder liners are kitted with either a service shim or captive shim that is 0.8128 mm [0.032 inch] thick. The liner has been cutback to accommodate this shim. Therefore the use of this shim is required. On blocks that have been machined, a second service shim will need to be stacked to account for the block machined depth. Block machining is **only** permitted to a maximum of 0.8128 mm [0.032 inch] so shims can **not** continue to be stacked to allow deeper block machining.

Note : If pistons are changed from high to low top ring pistons, the cylinder liners cannot be reused. This piston change creates a ring spacing difference in the cylinder liner wear pattern and can lead to ring seating and sealing issues. For piston

identification, refer to ISX15 and QSX15 Piston and Cylinder Liner Change History, Identification and Compatibility, Bulletin 4383774. (/qs3/pubsys2/xml/en/bulletin/4383774.html)

⚠ CAUTION ⚠

Antipolishing ring cylinder liners can only be used in combination with low top ring pistons. For more information on identifying a low top ring piston, refer to ISX15 and QSX15 Piston and Cylinder Liner Change History, Identification and Compatibility, Bulletin 4383774. (/qs3/pubsys2/xml/en/bulletin/4383774.html) If the antipolishing ring liner is used with a high top ring piston, it will result in a mechanical interference that will not allow engine reassembly.



Antipolishing ring cylinder liners integrate a removable carbon scraper ring known as the antipolishing ring into the top of the cylinder liner. The cylinder liner has a counterbore machined into the top where the antipolishing ring installs.

Once installed, the antipolishing ring overhangs the liner bore. This overhang keeps carbon scraped back on the piston top land. It is designed to limit carbon thickness so it does **not** contact the cylinder liner walls during operation. This prevents bore polish caused by piston carbon packing.

The antipolishing ring is symmetrical so there is no top or bottom distinction.

The antipolishing ring is serviceable and can be replaced without replacing the cylinder liner.

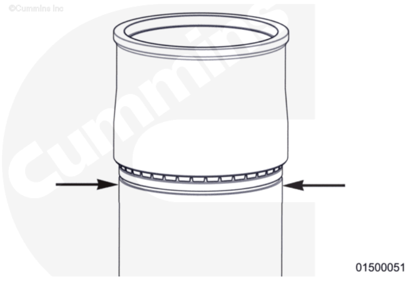
Antipolishing ring cylinder liners are flat top cylinder liners.

Antipolishing ring liners are only compatible with low top ring pistons. For more information, refer to ISX15 and QSX15 Piston and Cylinder Liner Change History, Identification and Compatibility, Bulletin 4383774. (/qs3/pubsys2/xml/en/bulletin/4383774.html)

The antipolishing ring liner has two bands machined around the bottom outside diameter of the liner. This indicates the liner is equipped with an antipolishing ring and a brass shim. These bands are intended for liner identification in the installed state by accessing the crankcase and viewing/feeling the bottom of the liner.

⚠ CAUTION ⚠

A large outside diameter liner will not install into a small cylinder bore block; however, the small outside diameter liner will install into a large bore block and may result in malfunction.



There are two different outside diameter sizes available. It is important to measure and understand the proper liner size for each engine.

The outside diameter of the cylinder liner needs to be measured in the packing area immediately below the o-ring. Use this measurement against the table below to determine proper replacement liner size.

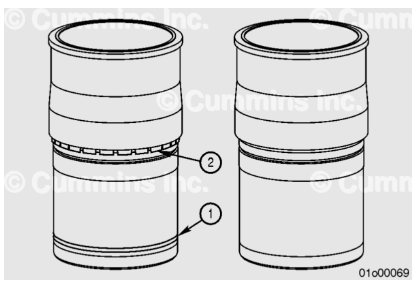
Note : The shimmed cylinder liner is backward compatible with all previous flat top liners and can be mixed or matched within an engine.

The following values should be used to determine block and liner compatibility and **not** as a reuse guideline. These are nominal values **only**, **not** specifications.

Liner Outside Diameter (At Packing Area)		Block Bore Diameter (Liner Packing Contact Area)	
Large Outside Diameter (OD) Liner		Large Cylinder Bore Block	
152 mm	[5.984 in]	152 mm	[5.984 in]
Small Outside Diameter (OD) Liner/Factory shimmed Liner		Small Cylinder Bore Block	
150 mm	[5.906 in]	150 mm	[5.906 in]

⚠ CAUTION ⚠

Liners kitted with shims must not be installed into a block without the shim. This will result in the cylinder liner being recessed into the block.



⚠ CAUTION ⚠

Once a factory shim has been removed from a cylinder liner, it can not be reused. The used shim must be discarded and a new factory shim installed.

⚠ CAUTION ⚠

Engines equipped with a factory shimmed cylinder liner assembly can not be repaired with a non-shimmed cylinder liner.

Liners with Hardened Brass Shims

Small Outside Diameter (150 mm [5.906 inch]) Liners with Factory Shim:

- The factory shimmed cylinder liners can easily be identified by the one or two bands machined in the lower outside diameter of the cylinder liner. This band is visible when the cylinder liner is installed. These cylinder liners incorporate a retained, hardened brass shim (2) at the mid-stop seating area. This is the **only** shim with retaining tabs stamped into it.

Large Outside Diameter (152 mm [5.984 inch]) Liners with Service Shim:

- The 152 mm [5.984 inch] cylinder liner requiring a service shim can easily be identified by one or two bands (1) machined in the lower outside diameter of the cylinder liner. These bands are intended for cylinder liner identification in the installed state by accessing the crankcase and viewing/feeling the bottom of the cylinder liner.

⚠ CAUTION ⚠

All six cylinder liners must be either the groove top or flat top design and the same outside diameter.

Flat Top vs. Groove Top Liner

Groove Top Cylinder Liner

- Legacy liner design used on early 15L engines: Has a groove machined into the top of the liner and is used with the multi-piece head gaskets.

Flat Top Cylinder Liner

- Current liner design: The top of the liner fully protrudes from the block. This liner does **not** have any grooves and is used with the current product single piece head gasket.

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50° C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- For power cylinder component identification, refer to ISX15 and QSX15 Piston and Cylinder Liner Change History, Identification and Compatibility, Bulletin 4383774. (/qs3/pubsys2/xml/en/bulletin/4383774.html)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html).
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/10/10-007-

037-tr.html)

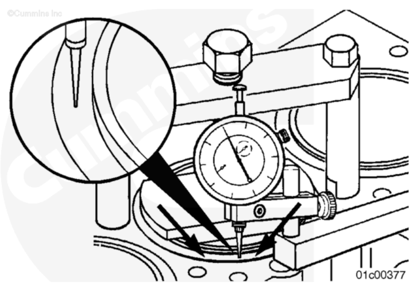
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/10/10-007-035-tr.html)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-089-tr.html)
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-046-tr.html)
- Remove the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-054-tr.html)

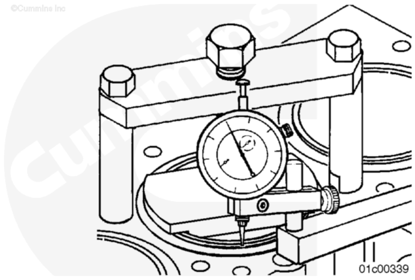
Note : If only liner protrusion is being measured and the liners are not being removed, the oil pan, oil suction tube, block stiffener plate, piston cooling nozzles, and piston and connection rod assembles do not need to be removed.

Initial Check

Liner Protrusion

For flat top cylinder liners, a protrusion measurement is taken, which is the height of the cylinder liner flange above the cylinder block surface, regardless of the flange or fire dam height.





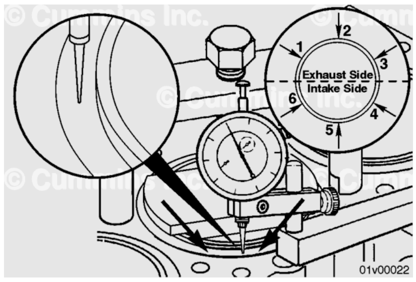
The cylinder liner measurement **must** be made with the cylinder liners in the clamped state.

For antipolishing ring liners, the antipolishing ring should remain installed for centering of the liner installation tool force plate.

Install and tighten the cylinder liner installation tool, Part Number 3164606, to clamp.

Rotate the force plate until the areas where the protrusion measurements will be taken are exposed.

Torque Value: 136 n•m [100 ft-lb]



Flat Top Cylinder Liners

- Use depth gauge, Part Number 3164438, to measure protrusion in the clamped state.

Note : Measure all six cylinders before removing any of the cylinder liners.

Individual Cylinder Measurements:

Single Cylinder Protrusion (measurement difference around individual cylinder):		
mm		in
0.0381	MAX	0.0015

Cylinder Liner Protrusion (around individual cylinder):		
mm		in
0.18	MIN	0.007
0.30	MAX	0.012

For each cylinder take a total of six measurements, three on the exhaust side, and three on the intake side. If any one of the individual measurements differs from the other five measurements by more than the specification listed above, or is outside the min/max specification listed above, reference the repair direction below.

Use the measurements taken above to average all six readings for each cylinder. Apply the averaged reading to the specifications listed below:

Cylinder to Cylinder Measurements:

- The difference between the lowest cylinder liner and the highest cylinder liner can **not** be greater than 0.102 mm

[0.004 in].
- The maximum allowable difference between adjacent cylinders is 0.051 mm [0.002 in].

If the averaged cylinder liner measurements are **not** within specifications listed above, reference the repair direction below:

Repair Direction:

- Remove all liners. Inspect all cylinder block counterbore shims and counterbore ledges for fretting and wear. Refer to the Cylinder Liner Counterbore Ledge Reuse Guidelines, Bulletin 4383753, (/qs3/pubsys2/xml/en/bulletin/4383753.html) to inspect the counterbore ledge surface areas for signs of pitting, fretting, or wear.
- For engines originally built without liner shims:
  - All six cylinder midstops **must** be machined and new liners containing shims **must** be install on all cylinder regardless of midstop condition.
- For engines which were originally built with liner shims:
  - Inspect the midstop for fretting and machine **only** cylinders which show evidence of fretting. A cylinder which does **not** show evidence of fretting does **not** need to be machined.
- Refer to Procedure 001-058 in Section (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)1.
  - The cylinder liner **must** be replaced for all cylinders that were machined.
  - If no fretting or wear is present, install new liners and measure liner protrusion again.
  - If protrusion is measured outside the specification, the counterbore **must** be machined.
  - If protrusion is measured inside the specification, proceed with the repair.

Note : The retained factory brass shim can not be used in place of the service shim for an engine with the cylinder block counterbores machined.

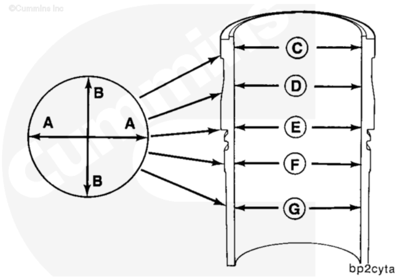
Liner Roundness

- While the liner is still installed, measure the liner bore for out-of-roundness at points "C," "D," "E," "F," and "G."
  - For antipolishing ring liners, take measurement C immediately below the antipolishing ring

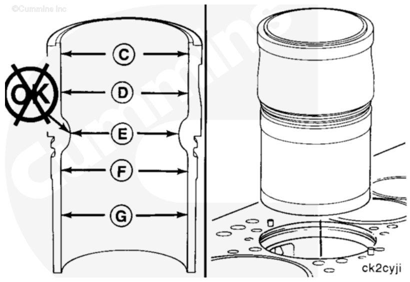
counterbore.

- Measure each point in the direction "AA" and "BB".
- The bore **must not** exceed the out-of-round specification.

Maximum Allowable Cylinder Liner Out-Of-Round		
mm		in
0.051	MAX	0.002



- If the liner bore is more than the out-of-round specification, remove the liner so the cylinder block liner bore roundness can be measured.
- The block counterbore diameters above and below the cylinder block counterbore area are **not** critical dimensions and do **not** need to be measured.

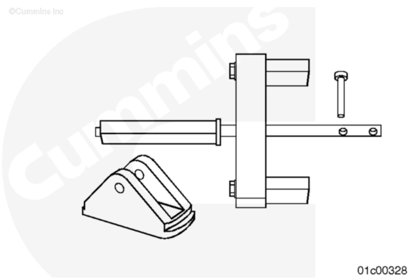


Remove

For antipolishing ring liners, removal of the antipolishing ring is necessary before removing the piston kits. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-054-tr.html)

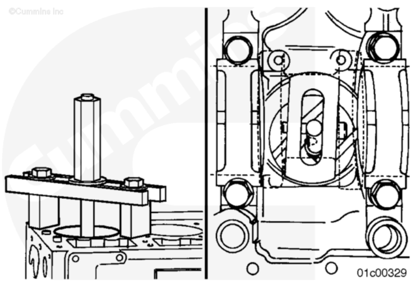
The cylinder liners can be removed by using either:

- Cylinder liner puller (universal), Part Number 3163745, and puller plate (S600), Part Number 3162462.
- Cylinder liner puller (universal), Part Number 3376015.



⚠ CAUTION ⚠

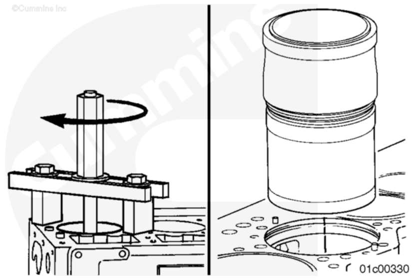
The cylinder puller must be installed and used as described to reduce the possibility of damage to the cylinder block. The puller plate must be parallel to the main bearing saddles and must not overlap the cylinder liner outside diameter.



Insert the cylinder liner puller onto the top of the cylinder block.

⚠ CAUTION ⚠

Once a factory shim has been removed from a cylinder liner, it can not be reused. The used shim must be discarded and a new factory shim installed.



The cylinder liner puller **must** be centered on the top of the cylinder block.

Turn the puller jackscrew **clockwise** to loosen the cylinder liner from the cylinder block.

Use both hands to remove the cylinder liner.

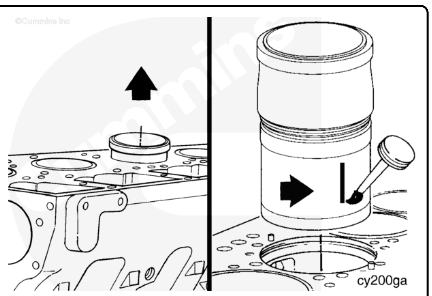
Remove and discard the o-ring seals.

If equipped, remove and discard the retained factory brass shim.

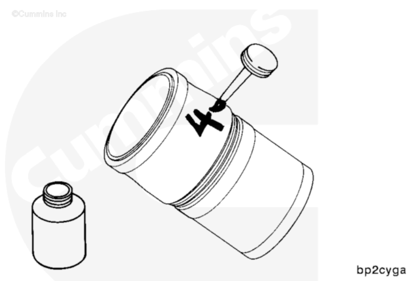
- Gently bend out the retention tabs and slide the shim off the cylinder liner. Discard the shim.

Note : If the engine previously had the counterbore mid-stop surface machined, the non-retained service shim must be removed, measured, and discarded. Shim thickness measurements are required to make sure the correct thickness service shim is installed on reassembly.

Use Dykem™, or equivalent, to place a mark on the intake air side of the cylinder liner to show cylinder liner orientation.



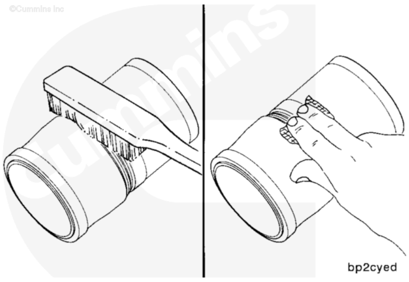
Use Dykem™, or equivalent, to mark the cylinder number on each cylinder liner.



Clean and Inspect for Reuse

⚠ CAUTION ⚠

Do not use emery cloth or sandpaper to remove carbon from the cylinder liners. Aluminum oxide or silicon particles from emery cloth or sand paper can cause serious engine damage. Do not use any abrasives in the ring travel area. The cylinder liner can be damaged.



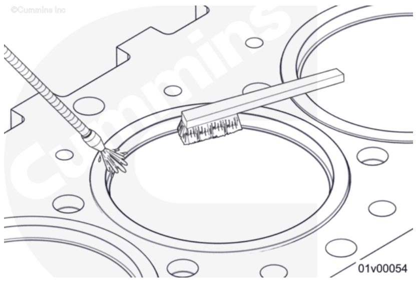
Note : If pistons are changed from high to low top ring pistons, the cylinder liners cannot be reused. This piston change creates a ring spacing difference in the cylinder liner wear pattern and can lead to ring seating and sealing issues. For piston identification, refer to ISX15 and QSX15 Piston and Cylinder Liner Change History, Identification and Compatibility, Bulletin 4383774. (/qs3/pubsys2/xml/en/bulletin/4383774.html)

Use a soft wire brush to clean the flange seating area.

Use a fine, fibrous, abrasive pad such as Scotch-Brite™ 7448, Part Number 3823258, or equivalent, to remove the remaining scale and rust.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Antipolishing Ring Cylinder Liner

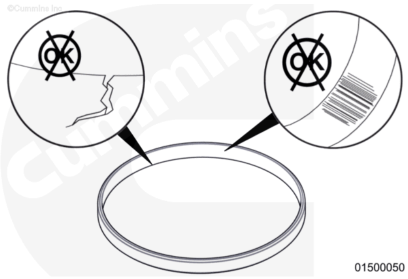
Inspect the antipolishing ring liner counterbore for carbon or debris that can interfere with seating of the antipolishing ring during final assembly.

Clean the counterbore. Use carbon cleaning solvent, Part Number 5298527. A brass wire brush can be used to help clean the carbon deposits.

Inspect the antipolishing ring counterbore for cracks or fretting. If cracks or fretting is found, the cylinder liner **must** be replaced.

Antipolishing Ring Cylinder Liner

Note : Do not install a used antipolishing ring into a new cylinder liner.



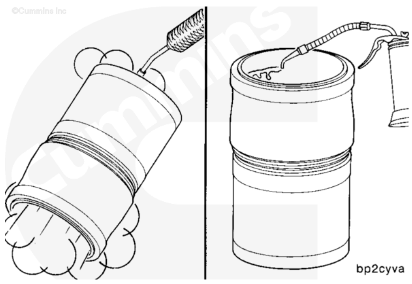
If the liner does **not** require replacement, inspect the antipolishing ring for cracks or fretting.

Discard the antipolishing ring and install a new ring if any damage is found.

Clean carbon from the antipolishing ring. Use carbon cleaning solvent, Part Number 5298527. A brass wire brush can be used to help clean the carbon deposits.

The antipolishing ring outside diameter should meet the following specification:

Antipolishing Ring Outside Diameter Specification		
mm		in
141.735	MIN	5.580
141.765	MAX	5.581



⚠ WARNING ⚠

Wear safety glasses or a face shield, as well as protective clothing, to reduce the possibility of personal injury when using a steam cleaner or high pressure water.

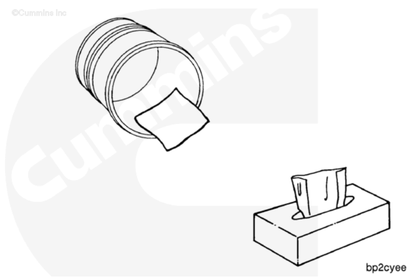
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent or steam clean the cylinder liners and dry with compressed air.

Use clean 15W-40 oil to lubricate the inside circumference of the cylinder liners.

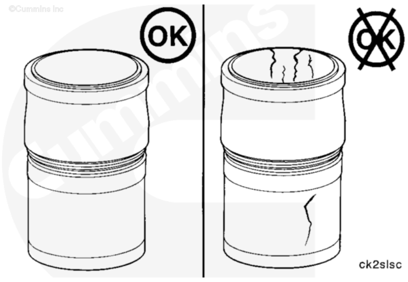
Allow the oil to soak into the cylinder liner for 5 to 10 minutes.



Use lint-free paper towels to wipe the oil from the inside of the cylinder liners.

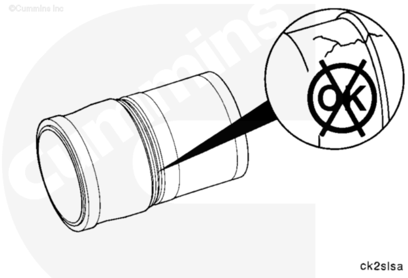
Continue to lubricate the inside of the cylinder liners and wipe clean until the paper towel shows no gray or black residue.

Inspect the cylinder liners for cracks on the inside and outside circumferences.



Inspect for cracks or fretting under the cylinder liner flange.

Cracks can also be detected by using either magnetic inspection or the dye method.

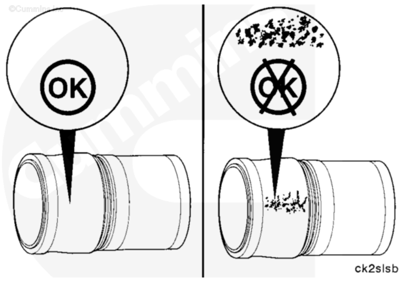


Inspect the outside circumference for excessive corrosion or pitting.

Cylinder liners with pitting generally can **not** be used again.

However, if the pitting is light and can be removed with fine emery cloth, the cylinder liner can be reused.

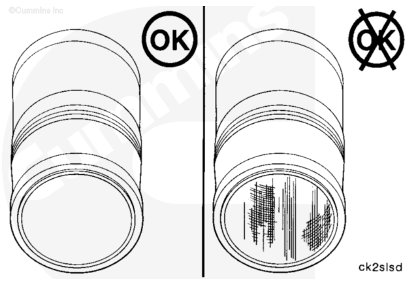
Cylinder Liner Pitting Depth		
mm		in
1.60	MAX	0.060



Inspect the inside circumference for vertical scratches deep enough to be felt with a fingernail.

If a fingernail catches in the scratch, the cylinder liner **must** be replaced.

Inspect the inside circumference for scuffing or scoring.

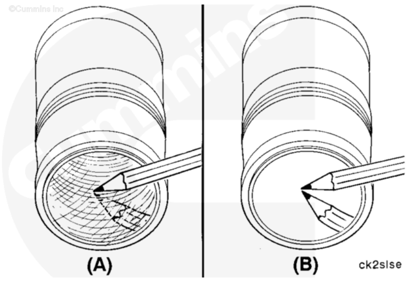


Inspect the inside diameter for cylinder liner bore polishing.

A moderate polish (A) produces a mirror finish in the worn area with traces of the original hone marks or an indication of an etch pattern.

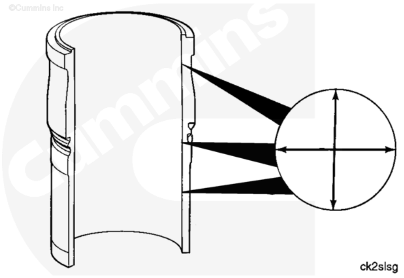
A heavy polish (B) produces a bright mirror finish in the worn area with no traces of hone marks or an etch pattern.

Replace the cylinder liner if a heavy polish is present.



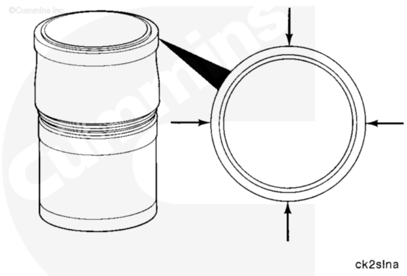
Use a dial bore gauge to measure the cylinder liner inside diameter in four places, 90 degrees apart at the top and bottom of the piston travel area.

Cylinder Liner Inside Diameter		
mm		in
137.14	MAX	5.40



Measure the cylinder liner top press fit area outside diameter.

Cylinder Top Press Fit Outside Diameter		
mm		in
160.98	MAX	6.34



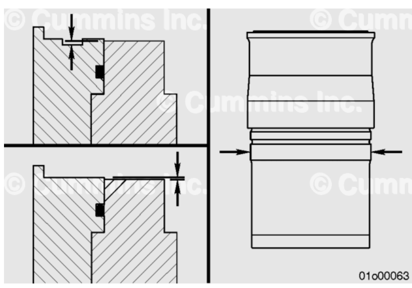
Install

⚠ CAUTION ⚠

All six cylinder liners must be either groove top or flat top design and the same outside diameter.

⚠ CAUTION ⚠

A large outside diameter liner will not install into a small cylinder bore block; however, the small outside diameter liner will install into a large bore block and may result in malfunction.



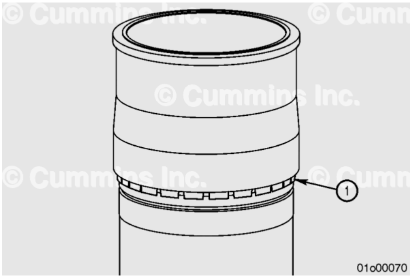
See the General Information step of this procedure for help with identification of a particular cylinder liner.

⚠ CAUTION ⚠

Engines equipped with a factory shimmed cylinder liner assembly can not be repaired with a non-shimmed cylinder liner.

⚠ CAUTION ⚠

Factory shimmed liners must not be installed into an engine without a shim present, as this will result in the cylinder liner being recessed into the block.



Make sure the cylinder block and all parts are clean before assembly.

Note : Do not reuse shims. Make sure to install new shims prior to installation.

If the engine was equipped with factory shimmed liners, new factory shims (1) **must** be installed.

- Install factory shims with retention tangs facing toward the bottom of the liner. Slide the shim up from the bottom of the liner until the retention tangs lock into the retention groove.

If the engine was equipped with service shims or the counterbores were machined during this repair event, new service shims of the correct thickness **must** be installed.

- Install the service shim into the cylinder liner bore, centering the shim on the block mid-stop seat.

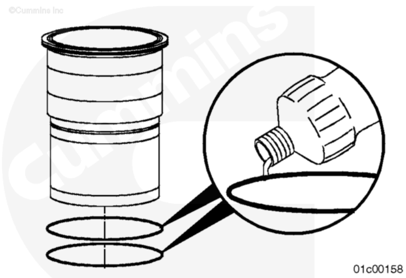
Note : Installing a factory shimmed cylinder liner assembly on top of a service shim is allowed on engines that have had the cylinder block counterbores machined.

Lubricate the cylinder liner o-ring seal with vegetable oil prior to assembly.

The cylinder liner **must** be installed within 15 minutes of lubricating the o-ring.

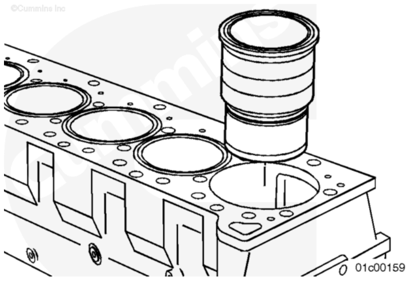
Install the o-ring seal onto the cylinder liner.

Be sure the o-ring is seated flat in the o-ring groove.



⚠ CAUTION ⚠

The fire dam is brittle. Care must be taken to prevent damage to the liner.



Install the cylinder liner into the cylinder block.

When acceptable used cylinder liners are installed, rotate the cylinder liner 90 degrees from its original position in the engine. The thrust and anti-thrust surfaces **must** face the front and back of the cylinder block.

Use Cylinder Liner Installation Kit, Part Number 3162461, to seat the liner.

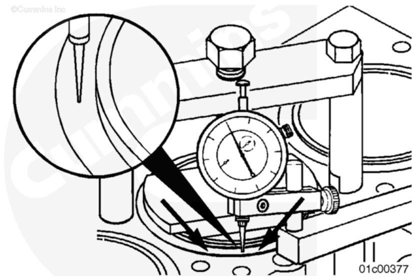
If the cylinder liner does **not** seat properly, remove it and inspect the cylinder liner bore seat and cylinder liner for nicks, burrs, dirt, or an out-of-place shim, if equipped.

Install the cylinder liner again.

Measure

Liner Protrusion

For flat top cylinder liners, a protrusion measurement is taken, which is the height of the cylinder liner flange above the cylinder block surface, regardless of the flange or fire dam height.

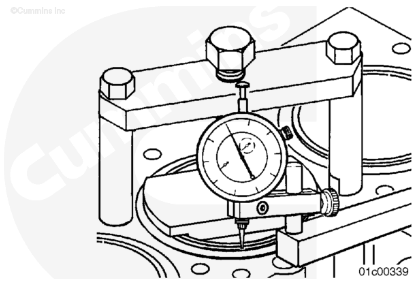


The cylinder liner measurement **must** be made with the cylinder liners in the clamped state.

For antipolishing ring liners, the antipolishing ring should remain installed for centering of the liner installation tool force plate.

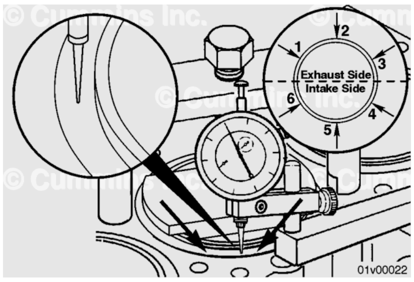
Install and tighten the cylinder liner installation tool, Part Number 3164606, to clamp.

Rotate the force plate until the areas where the protrusion measurements will be taken are exposed.



Torque Value: 136 n•m [100 ft-lb]

Flat Top Cylinder Liners



- Use depth gauge, Part Number 3164438, to measure protrusion in the clamped state.

Note : Measure all six cylinders before removing any of the cylinder liners.

Individual Cylinder Measurements:

Single Cylinder Protrusion (measurement difference around individual cylinder):		
mm		in
0.0381	MAX	0.0015

Cylinder Liner Protrusion (around individual cylinder):		
mm		in
0.18	MIN	0.007
0.30	MAX	0.012

For each cylinder take a total of six measurements, three on the exhaust side, and three on the intake side. If any one of the individual measurements differs from the other five measurements by more than the specification listed above, or is outside the min/max specification listed above, reference the repair direction below.

Use the measurements taken above to average all six readings for each cylinder. Apply the averaged reading to the specifications listed below:

Cylinder to Cylinder Measurements:

- The difference between the lowest cylinder liner and the highest cylinder liner can **not** be greater than 0.102 mm [0.004 in].
- The maximum allowable difference between adjacent cylinders is 0.051 mm [0.002 in].

If the averaged cylinder liner measurements are **not** within specifications listed above, reference the repair direction below:

Repair Direction:

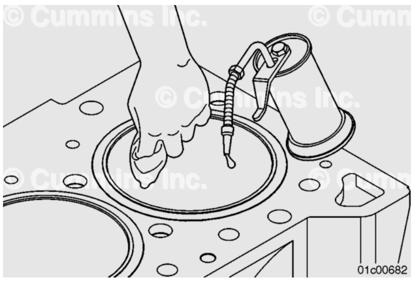
- Remove all liners. Inspect all cylinder block counterbore shims and counterbore ledges for fretting and wear. Refer to the Cylinder Liner Counterbore Ledge Reuse Guidelines, Bulletin 4383753, (/qs3/pubsys2/xml/en/bulletin/4383753.html) to inspect the counterbore ledge surface areas for signs of pitting, fretting, or wear.

- If fretting or wear is present on any cylinder, all six cylinders **must** be machined. Refer to Procedure 001-058 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-058-tr.html)
  - The cylinder liner **must** be replaced for all cylinders that were machined.
- If no fretting or wear is present, install new liners and re-measure liner protrusion.
  - If protrusion is measured outside the specification, the counterbore **must** be machined.
  - If protrusion is measure inside the specification, proceed with the repair.

Note : The retained factory brass shim can not be used in place of the service shim for an engine with the cylinder block counterbores machined.

Remove the liner installation tool.

Before installing the piston and connecting rod assemblies, wipe the inside of the cylinder liners. Use a clean cloth with 15W-40 oil.



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

Note : All engines **must** be run-in after a rebuild or a repair involving the replacement of one or more piston ring sets, cylinder liners, or pistons. Use the following procedure for a general run-in test overview. Refer to Procedure 014-999 in section F. (/qs3/pubsys2/xml/en/procedures/10/10-014-999.html)

- Install the piston and connecting rod assemblies. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-054-tr.html)
- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-046-tr.html)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1. (/qs3/pubsys2/xml/en/procedures/10/10-001-089-tr.html)
- Install the oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/10/10-007-035-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/10/10-007-025-tr.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/10/10-002-004-tr.html)
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html)
- Prime the lubricating oil system. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/10/10-007-037-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
- Operate the engine. Check for leaks.